

A PROJECT TITLE
ON
MULTI-CLASS MATCH RESULT PREDICTION USING MULTINOMIAL
LOGISTIC REGRESSION

project report submitted in partial fulfilment of the requirements for the Course

MACHINE LEARNING
BACHELOR OF TECHNOLOGY

in

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by

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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING



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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

Name of Title:

Multi-Class Match Result Prediction Using Multinomial Logistic Regression

1. Abstract:

This project focuses on predicting the result of a sports match, whether it ends in a win, loss, or draw, using a machine learning technique called Multinomial Logistic Regression. Most real-world matches, especially in team sports like football, can end in more than two outcomes, so a simple binary model is not enough. Multinomial Logistic Regression solves this problem by handling three or more outcome classes at the same time, making it a good fit for match result prediction. The project begins by collecting historical match data, which includes information such as each team recent form, whether the team is playing at home or away, past head-to-head results, and average goals scored and conceded. This data is cleaned and organized so it can be used to train the model. Once ready, the data is split into training and testing sets, and the model is trained to learn the relationship between these features and the final match outcome. After training, the model is tested on new data to see how accurately it can predict future results. Its performance is measured using simple metrics such as accuracy, precision, recall, and a confusion matrix, which show how well the model separates wins, losses, and draws. One of the biggest advantages of this approach is that it is easy to understand and interpret, unlike many complex machine learning models. This allows analysts, fans, and sports platforms to see which factors, such as home advantage or recent form, have the greatest impact on the outcome of a match. The final goal of this project is to build a working, easy-to-use prediction system that can support sports analysis, fan engagement platforms, and match previews. By using a simple and interpretable model like Multinomial Logistic Regression, the project shows that useful and fairly accurate sports predictions can be made without relying on very complex machine learning techniques.

2. INTRODUCTION:

Predicting match outcomes is a popular problem in the sports and entertainment industry, useful for fans, analysts, and broadcasters. Most matches can end in three ways, win, loss, or draw, so this is treated as a multi-class classification problem. Multinomial Logistic Regression is used because it is simple, easy to interpret, and works well for problems with more than two outcomes. The model uses features like team form, home advantage, and past performance to predict results. This project aims to build such a model, train it on match data, test its accuracy, and understand which factors matter most in deciding the result.

3. SOFTWARE REQUIREMENTS:

Component	Requirement
Operating System	Windows / Linux / macOS
Programming Language	Python 3.9 or above
IDE	Jupyter Notebook / VS Code / Google Colab
Libraries	pandas, numpy, scikit-learn, matplotlib


 Signature of the Guide


 Course Instructor